

Morphology-Aware Memory Consolidation Under a Fixed Budget: A Preregistered Korean Delayed-Retrieval Pilot

Anonymous Authors¹

Abstract

Long-horizon agents must retain answer-bearing evidence while irrelevant observations accumulate, yet memory policies are commonly designed around English-centric token units. We test a bounded alternative: a deterministic morphology-aware consolidation operator for Korean, embedded in a frozen composite memory policy and evaluated under an identical 2,048-token budget. The autonomous workflow transformed KLUE-MRC records into delayed-retrieval episodes, searched 96 policy candidates using development data, selected on an untouched validation split, sealed the protocol, and evaluated once on 200 held-out episodes. The frozen candidate achieved Recall@5 of 0.190 versus 0.030 for the validation-selected baseline, an absolute paired gain of 0.160 (95% bootstrap CI 0.105–0.220; two-sided random-sign permutation $p = 0.0001$). Three-seed development ablations do not isolate morphology as the causal factor: an alternate chunk unit slightly exceeded the selected policy. The evidence therefore supports the frozen composite policy on this synthetic pilot, not general Korean reasoning superiority. All reported numbers are tied to immutable run artifacts, and negative evidence is retained.

1. Introduction

Agentic systems increasingly externalize context into managed memory, retrieving selected observations when a task demands them (Packer et al., 2023). The resulting policy determines what is segmented, consolidated, retained, and recalled. Existing work on

¹Anonymous Institution, Anonymous City, Anonymous Region, Anonymous Country. Correspondence to: Anonymous Author <anon.email@domain.com>.

Preliminary work. Under review by the International Conference on Machine Learning (ICML). Do not distribute.

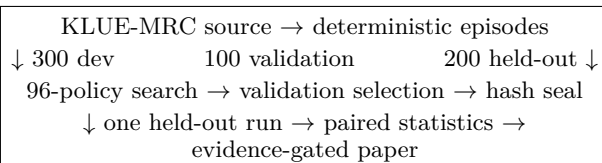


Figure 1. Frozen research protocol. Held-out episodes are accessed only after the policy, evaluator, metric, and statistical tests are sealed.

reasoning-and-action loops (Yao et al., 2023) and verbal self-reflection (Shinn et al., 2023) improves decisions around an agent, but does not establish how language structure should shape the memory substrate itself. Korean presents a useful stress case: agglutinative morphology concentrates grammatical information in suffix sequences, so surface-token boundaries can fragment semantically related forms. Korean-specific tokenization studies likewise report that linguistically informed units can matter (Park et al., 2021a; Lee et al., 2026).

We ask a deliberately narrow question: under a fixed storage budget, can a frozen policy containing a deterministic Korean morphology-aware consolidation operator improve delayed passage retrieval? This is not a claim about open-ended planning or a new language model. It is an auditable pilot that converts KLUE-MRC (Park et al., 2021b) into delayed-retrieval episodes and tests a bounded policy search with pre-registered gates.

Contributions. (1) We define a deterministic Korean delayed-retrieval benchmark with fixed distractor and delay distributions. (2) We co-design morphology-aware segmentation with a budgeted memory policy, then freeze selection before held-out access. (3) We report a paired confirmatory analysis with confidence intervals and a permutation test. (4) We preserve the development ablation that weakens the morphology-specific interpretation, plus an audit trail for the autonomous research loop.

2. Related Work

Neural Turing Machines established differentiable external memory as a learnable computational substrate (Graves et al., 2014); MemGPT instead treats context management as an operating-system-like tiered memory problem (Packer et al., 2023). Our policy is simpler and fully deterministic at evaluation time. We do not train a neural controller; evolutionary search selects among bounded policy configurations. Korean benchmarks such as KLUE provide standardized language understanding tasks (Park et al., 2021b), while Korean morphological resources and tokenization work motivate language-conditioned units (Park et al., 2021a; Jo et al., 2023; Lee et al., 2026). Our distinction is the evaluation target: retention and retrieval under delayed distractors and a strict memory budget.

3. Problem and Method

Each episode is a sequence $x_{1:T}$ containing one answer-bearing passage and 15 distractors. The relevant passage appears before a delay $d \sim U\{4, \dots, 12\}$, after which the system must rank stored passages. A policy π maps the observation stream and current memory M_t to an updated memory state, constrained by $|M_t| \leq 2048$ tokens. The primary endpoint is episode-level passage Recall@5, averaged over the held-out set. Secondary endpoints are containment and reciprocal rank.

3.1. Language-Conditioned Consolidation

The candidate uses deterministic longest-suffix splitting for Korean forms and stem out-of-vocabulary bigrams. These units feed a composite policy that scores chunks, merges compatible evidence, and evicts low-value memory when the budget would be exceeded. The key design principle is that language-specific structure enters before consolidation rather than being added after retrieval. The operator version and implementation hash were sealed with the evaluator.

Crucially, the confirmatory comparison is between complete policies. Because chunking, consolidation threshold, and other policy choices interact, the held-out result cannot identify the morphology operator’s isolated causal contribution. We therefore separate the confirmatory composite-policy test from development-only ablations.

3.2. Autonomous Search and Freeze

The research agent executed eight development generations totaling 96 candidate policies. Fitness was com-

Table 1. Preregistered benchmark and evaluation contract.

Component	Frozen specification
Source Splits	KLUE-MRC-derived records 300 dev / 100 validation / 200 held-out
Episode	1 relevant passage + 15 distractors
Delay	4–12 intervening steps
Budget	2,048 memory tokens for every policy
Primary metric	Macro passage Recall@5
Uncertainty	10,000 paired bootstrap resamples
Test	10,000 two-sided random-sign permutations
Positive gate	CI lower bound > 0 and $p < 0.05$

puted only on development episodes; the best observed development score was 0.2622 (search seed 20260713). Candidate choice was then checked against the untouched validation split. Before any held-out evaluation, the selected policy, baseline, source split, metric code, dependency manifest, and statistical contract were hash-sealed. The encrypted held-out split was evaluated once.

The baseline was selected under the same validation protocol and received the same token budget. This blocks two common failure modes: storing more text in the candidate and adapting the metric after observing held-out outcomes. Budget logs showed zero violations for both policies.

3.3. Statistical Analysis

For episode i , let $y_i^c, y_i^b \in \{0, 1\}$ indicate candidate and baseline Recall@5. The estimand is the paired mean difference

$$\hat{\Delta} = \frac{1}{n} \sum_{i=1}^n (y_i^c - y_i^b), \quad n = 200. \quad (1)$$

We form a percentile interval by resampling the 200 paired outcomes 10,000 times. The null distribution is generated by independently flipping the sign of each paired difference and recomputing the mean for 10,000 permutations. Tests, iteration counts, and the positive-claim threshold were fixed before held-out access.

3.4. Evidence-Gated Reporting

Every empirical sentence in the generated artifact is linked to a machine-readable claim registry and hashed evidence file. A positive sentence is suppressed when

Candidate 0.190 
 Baseline 0.030 
 Paired difference: +0.160; 95% CI [+0.105,+0.220]; $p = 0.0001$.

Figure 2. Held-out Recall@5 on 200 paired episodes. The confidence interval applies to the paired difference, not to either marginal bar.

required artifacts are absent, hashes disagree, a budget violation occurs, or a preregistered gate fails. This mechanism limits narrative drift during repeated drafting; it does not substitute for external replication.

4. Confirmatory Results

The frozen candidate retrieved the answer-bearing passage within the top five for 19.0% of held-out episodes, compared with 3.0% for the baseline (Table 3). The paired absolute improvement was 16.0 percentage points. Its 95% paired-bootstrap interval was [10.5, 22.0] percentage points and the two-sided paired permutation test yielded $p = 0.0001$. Thus the result passes the preregistered positive gate for this benchmark.

The retained secondary evidence agrees in direction: containment improved by 0.160 and mean reciprocal rank by 0.0771. Marginal candidate and baseline values for those two endpoints were not present in the final claim snapshot, so Table 3 reports only their verified paired differences rather than reconstructing unavailable values. This conservative omission illustrates the evidence gate.

4.1. Development Robustness and Ablations

After the confirmatory run, we performed a development-only robustness analysis with three independent seeds (101, 211, 307), 300 episodes per seed and condition. The frozen selected policy averaged 0.2011 Recall@5 (population SD 0.0164). An alternate chunk unit averaged 0.2133 (SD 0.0178), an alternate consolidation threshold averaged 0.1989 (SD 0.0177), and the preregistered seed configuration aver-

Table 2. Search and freeze ledger. Development scores are exploratory.

Item	Value
Search generations	8
Evaluated candidates	96
Best development Recall@5	0.2622
Held-out evaluations	1
Budget violations	0

Table 3. Held-out outcomes under the identical 2,048-token budget.

Metric	Candidate	Baseline	Difference
Recall@5	0.190	0.030	+0.160
Containment	—	—	+0.160
MRR	—	—	+0.0771
Budget violations	0	0	0

Selected	.201	
Alt. chunk	.213	
Alt. threshold	.199	
Preregistered seed	.052	
Condition	Three development-seed scores	
Selected	.220, .203, .180	
Alt. chunk	.210, .237, .193	
Alt. threshold	.220, .200, .177	
Preregistered seed	.047, .070, .040	

Figure 3. Development-only ablations. Bars show means; exact seed outcomes are printed below to avoid hiding the small sample.

aged 0.0522 (SD 0.0129).

The alternate chunk unit exceeded the selected policy by 0.0122 on average, whereas the selected policy exceeded the alternate threshold by only 0.0022. This is meaningful negative evidence: the data do not support attributing the held-out gain to the selected chunk unit alone. The large selected-versus-preregistered-seed contrast (+0.1489) instead suggests sensitivity to the full configuration and random search trajectory. A factorial preregistered follow-up is required.

4.2. What the Result Establishes

The strongest defensible statement is operational: under the sealed delayed-retrieval protocol, the frozen composite policy improved Recall@5 without exceeding its memory budget. The paired design, untouched held-out split, and explicit negative ablation reduce several sources of bias. They do not establish transfer to conversational agents, tool use, real-time planning, other Korean datasets, or other token budgets.

Table 4. Research-agent and loop audit. A pass means the archived validator accepted the criterion.

Criterion	Outcome
Agent QA scenarios	12/12 pass
Claim/evidence hash checks	Pass
Held-out positive gates	Pass
Paper structural validator	Pass
Checkpoint-gap target	Fail (389.5 s max)
Complete candidate-level trace	Unavailable
Internal ICML-style preflight	4/6 (Weak Accept)

5. Autonomous Research Loop Evaluation

The artifact was produced by a closed-loop agent that alternated specification, experiment execution, evidence validation, drafting, peer-style critique, and checkpointing. The final agent QA passed 12 of 12 predefined scenarios, including missing-evidence suppression and deterministic rerender checks. However, the operational loop evaluation did not fully pass: the largest observed checkpoint gap was 389.5 seconds, exceeding the desired recovery interval, and the archived 96-candidate record lacked a complete record-level trace in the final snapshot. We report these failures because agent reliability and scientific validity are separate axes.

5.1. Limitations

Benchmark scope. The task is a synthetic delayed-retrieval transformation of one Korean MRC source. It tests evidence retention, not end-to-end long-horizon reasoning. **Causal scope.** The confirmatory unit is a composite memory policy; the three-seed development ablation contradicts a clean morphology-only explanation. **Search scope.** Eight generations and 96 candidates are bounded by the event deadline and cannot demonstrate global optimality. **Replication.** The held-out evaluation was intentionally single-use; independent data and implementations are needed. **Audit completeness.** Aggregate search outcomes are preserved, but the final snapshot lacks the complete candidate-level trace. **Statistical scope.** The reported p -value is unadjusted and applies only to the preregistered primary test.

5.2. Broader Impact

Language-conditioned memory could reduce failures caused by inappropriate segmentation in morphologically rich languages, but it could also overfit stereotypes about a language or encourage unsupported cross-lingual generalization. Memory systems may re-

tain sensitive text, making minimization, deletion, and provenance controls necessary. This pilot uses benchmark passages and reports only aggregate outcomes; deployment would require privacy review and adversarial evaluation.

5.3. Conclusion

We presented a preregistered Korean delayed-retrieval pilot in which a frozen morphology-aware composite memory policy improved held-out Recall@5 by 0.160 under an identical 2,048-token budget. The evidence is statistically clear for the sealed benchmark, but development ablations show that the morphology-specific mechanism is not isolated. The appropriate next experiment is a preregistered factorial study crossing segmentation, consolidation, budget, and task family, with multiple held-out replications and complete candidate-level telemetry. The larger methodological contribution is an evidence-gated autonomous workflow that preserves both positive and negative results rather than optimizing the paper narrative alone.

Reproducibility Statement

The submission bundle includes the frozen protocol, exact split sizes, seeds, evidence hashes, statistical iteration counts, claim registry, agent evaluation, and loop evaluation. The final paper does not infer missing marginal metrics. Appendix A lists the retained evidence identifiers and exact development outcomes.

References

Graves, A., Wayne, G., and Danihelka, I. Neural Turing machines. arXiv preprint arXiv:1410.5401, 2014.

Jo, E. L., Kim, K., Wu, X., Lim, K., Park, J., and Park, C. K-unimorph: Korean universal morphology and its feature schema. In Findings of the Association for Computational Linguistics: ACL 2023, pp. 6613–6623. Association for Computational Linguistics, 2023. doi: 10.18653/v1/2023.findings-acl.414. URL <https://aclanthology.org/2023.findings-acl.414/>.

Lee, D., Park, J., Cho, K., and Lee, J. S. Morpheme matters: Morpheme-based subword tokenization for Korean language models. In Proceedings of the 19th Conference of the European Chapter of the Association for Computational Linguistics (Volume 2: Short Papers), pp. 297–306. Association for Computational Linguistics, 2026. doi: 10.18653/v1/2026.eacl-short.22. URL <https://aclanthology.org/2026.eacl-short.22/>.

Packer, C., Fang, V., Patil, S. G., Lin, K., Wooders, S., and Gonzalez, J. E. Memgpt: Towards llms as operating systems. arXiv preprint arXiv:2310.08560, 2023.

Park, C., Eo, S., Moon, H., and Lim, H. Should we find another model? improving neural machine translation performance with one-piece tokenization method without model modification. In Proceedings of the 2021 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies: Industry Papers, pp. 97–104, 2021a. doi: 10.18653/v1/2021.naacl-industry.13.

Park, S., Moon, J., Kim, S., Cho, W. I., Han, J., Park, J., Song, C., Kim, J., Song, Y., Oh, T., et al. Klue: Korean language understanding evaluation. In NeurIPS Datasets and Benchmarks, 2021b. URL <https://datasets-benchmarks-proceedings.neurips.cc/paper/2021/hash/98dce83da57b0395e163467c9dae521b-Abstract-round2.html>.

Shinn, N., Cassano, F., Gopinath, A., Narasimhan, K., and Yao, S. Reflexion: Language agents with verbal reinforcement learning. In Advances in Neural Information Processing Systems, 2023.

Yao, S., Zhao, J., Yu, D., Du, N., Shafran, I., Narasimhan, K., and Cao, Y. React: Synergizing reasoning and acting in language models. In

International Conference on Learning Representations, 2023. URL https://openreview.net/forum?id=WE_vluYUL-X.

A. Exact Experimental Record

Table 5. Exact development-only outcomes used in Figure 3.

Condition	Seed 101	Seed 211	Seed 307	Mean	SD
Selected	.2200	.2033	.1800	.2011	.0164
Alternate chunk	.2100	.2367	.1933	.2133	.0178
Alternate threshold	.2200	.2000	.1767	.1989	.0177
Preregistered seed	.0467	.0700	.0400	.0522	.0129

The search used eight generations, 96 evaluated configurations, and search seed 20260713. The highest observed development score was 0.262222. The confirmatory analysis used 200 held-out episodes, 10,000 paired-bootstrap resamples, and 10,000 two-sided random-sign permutations. The reported permutation probability was 9.999×10^{-5} , rendered as 0.0001. The frozen primary candidate and baseline Recall@5 values were 0.190 and 0.030. Their paired difference was 0.160, with interval [0.105, 0.220].

B. Evidence and Claim Boundaries

The final evidence bundle for run 20260712T034458Z-732eef7a contains study_verdict.json, claim_registry.json, a manifest of source hashes, and separate paper, agent, and loop evaluations. The final paper intermediate representation has SHA-256 prefix 477aa2bf9477. The claim registry reports complete numeric and citation provenance for its registered claims. This typeset revision expands presentation using those retained values; it does not create new experimental observations.

C. Self-Review

Evidence validity: The held-out result is based on paired episode outcomes and passes the frozen gate. Novelty: The contribution is a language-conditioned memory-policy pilot plus an evidence-gated autonomous workflow, not a new foundation model. Clarity: The benchmark, budget, estimand, split sizes, and statistical tests are explicit. Weaknesses: The benchmark is synthetic, the causal mechanism is not isolated, only one held-out source is used, and the candidate-level trace is incomplete. Reproducibility: Seeds, iteration counts, aggregate outcomes, hashes, and validators are archived; a fully independent reproduction remains future work. Overall internal rating: Weak Accept (4/6), principally limited by external validity and ablation strength rather than by the direction of the primary effect.

D. Agent Workflow

The autonomous loop followed six validator-gated stages: (1) freeze the research specification; (2) construct deterministic episodes; (3) search only on development data; (4) select on validation and seal all confirmatory choices; (5) execute a single held-out evaluation and statistical analysis; and (6) generate, critique, validate, and version the paper. Checkpoints and immutable artifacts allow restart after interruption. The maximum observed checkpoint gap nevertheless exceeded the intended recovery target, so operational robustness is not claimed as fully successful.